PHYSICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

26. An electron is made to enter symmetrically between two parallel and equally but oppositely charged metal plates, each of 10 cm length. The electron emerge out of the electric field region with a horizontal component of velocity 10^6 m/s. If the magnitude of the electric field between the plates is 9.1 V/cm, then the vertical component of velocity of electron is

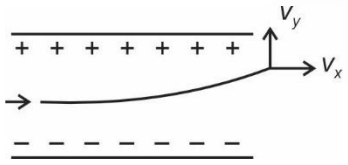
(mass of electron = 9.1×10^{-31} kg and charge of electron = 1.6×10^{-19} C)

(1) 0 (2) 16×10^6 m/s

(3) 16×10^4 m/s (4) 1×10^6 m/s

Answer (2)

Sol.



Horizontal component of velocity will remain same.

$$t = \frac{l}{v_x} = \frac{0.1}{10^6}$$

$$v_y = a_y t = \frac{eE}{m} t$$

$$= \frac{1.6\times10^{-19}\times9.1\times10^{2}}{9.1\times10^{-31}} \times \frac{0.1}{10^6}$$

$$= 1.6 \times 10^7$$

27. An amount of ice of mass 10^{-3} kg and temperature –10°C is transformed to vapour of temperature 110°C by applying heat. The total amount of work required for this conversion is,

(Take, specific heat of ice = 2100 $Jkg^{-1}K^{-1}$, specific heat of water 4180 $Jkg^{-1}K^{-1}$, specific heat of steam = 1920 $Jkg^{-1}K^{-1}$, Latent heat of ice = 3.35×10^5 Jkg^{-1} and Latent heat of steam = 2.25×10^6 Jkg^{-1})

(1) 3022 J

(2) 3043 J

(3) 3024 J

(4) 3003 J

Answer (2)

Sol. ice at –10°C ⟶ ice at 0°C ⟶ water at 0°C ↓ water at 100° ⟶ vapour at 100°C ⟶ vapour at 110°C

$$Q = 10^{-3}\left[2100\times10 + 3.35\times10^5 + 100\times4180 + 2.25\times10^6 + 10\times1920\right]$$

$$= 3043.2 \text{ J}$$

28. If B is magnetic field and μ_0 is permeability of free space, then the dimensions of (B/μ_0) is

(1) $ML^2T^{-2}A^{-1}$

(2) $L^{-1}A$

(3) $LT^{-2}A^{-1}$

(4) $MT^{-2}A^{-1}$

Answer (2)

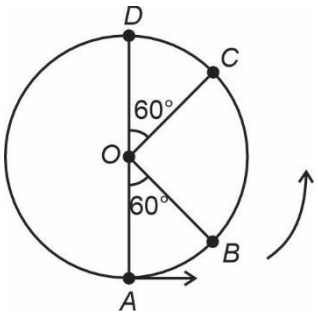
Sol. For a current carrying loop at centre

$$B = \frac{\mu_0 i}{2R}$$

$$\therefore \quad \frac{B}{\mu_0} \equiv \frac{i}{R} \equiv [AL^{-1}]$$



29. A bob of mass m is suspended at a point O by a light string of length l and left to perform vertical motion (circular) as shown in figure. Initially, by applying horizontal velocity v_0 at the point 'A', the string becomes slack when, the bob reaches at the point 'D'. The ratio of the kinetic energy of the bob at the point B and C is _____.



(1) 1

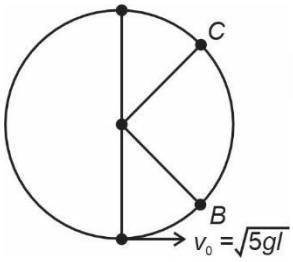
(2) 3

(3) 4

(4) 2

Answer (4)

Sol.



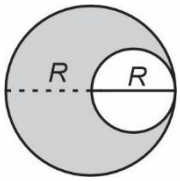
$$v_B = \sqrt{v_0^2 - 2g\left(\frac{l}{2}\right)}$$

$$v_B = \sqrt{4gl}$$

$$v_C = \sqrt{v_0^2 - 3gl} = \sqrt{2gl}$$

$$\frac{K_B}{K_C} = \frac{v_B^2}{v_C^2} = 2$$

30. A uniform circular disc of radius 'R' and mass 'M' is rotating about an axis perpendicular to its plane and passing through its centre. A small circular part of radius $R/2$ is removed from the original disc as shown in the figure. Find the moment of inertia of the remaining part of the original disc about the axis as given above.

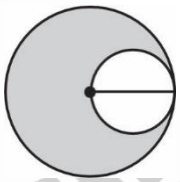


(1) $\frac{7}{32}MR^2$ (2) $\frac{17}{32}MR^2$

(3) $\frac{13}{32}MR^2$ (4) $\frac{9}{32}MR^2$

Answer (3)

Sol.



$$I_{complete} = \frac{MR^2}{2}$$

$$I_{removed} = \left(\frac{M}{4}\right)\left(\frac{R}{2}\right)^2\frac{1}{2} + \left(\frac{M}{4}\right)\left(\frac{R}{2}\right)^2$$

$$= \frac{3MR^2}{32}$$

$$I = \frac{MR^2}{2} - \frac{3MMR^2}{32} = \frac{13}{32}MR^2$$

31. A parallel-plate capacitor of capacitance 40μF is connected to a 100 V power supply. Now the intermediate space between the plates is filled with a dielectric material of dielectric constant K = 2. Due to the introduction of dielectric material, the extra charge and the change in the electrostatic energy in the capacitor, respectively, are

(1) 2 mC and 0.4 J (2) 8 mC and 2.0 J

(3) 4 mC and 0.2 J (4) 2 mC and 0.2 J



Answer (3)

Sol. $C = 40\ \mu F$

$C' = kC = 80\ \mu F$

$V = 100\ V$

$Q = CV = 4\ mC$

$Q' = C'V = 8\ mC$

$\therefore$ Extra charge = 4 mC

$$U_1 = \frac{1}{2}CV^2,\ U_2 = \frac{1}{2}C'V^2$$

$$\Delta U = \frac{1}{2}(C' - C)V^2$$

$$= \frac{1}{2} \times 40 \times 10^4 \times 10^{-6}$$

$$= 0.2\ J$$

32. The work functions of cesium (Cs) and lithium (Li) metals are 1.9 eV and 2.5 eV, respectively. If we incident a light of wavelength 550 nm on these two metal surfaces, then photo-electric effect is possible for the case of

(1) Neither Cs nor Li (2) Both Cs and Li

(3) Cs only (4) Li only

Answer (3)

Sol. E(Energy of incident photon) $= \frac{1240}{550} = 2.25$ eV

$E > \phi_{Cs}$, $E < \phi_{Li}$

$\therefore$ Only cesium will show photoelectric effect.

33. An electron in the ground state of the hydrogen atom has the orbital radius of 5.3×10^{-11} m while that for the electron in third excited state is 8.48×10^{-10} m. The ratio of the de Broglie wavelengths of electron in the ground state to that in the excited state is

(1) 4 (2) 16

(3) 3 (4) 9

Answer (*)

Sol. According to de-Broglie hypothesis

$$2\pi r_n = n\lambda$$

$$\frac{2\pi r_1}{2\pi r_4} = \frac{\lambda_1}{4\lambda_4}$$

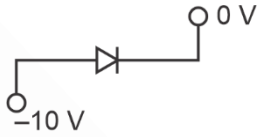
$$\frac{\lambda_1}{\lambda_4} = 4\left(\frac{r_1}{r_4}\right)$$

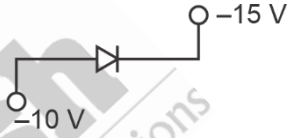
$$r \propto n^2$$

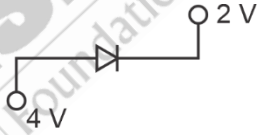
$$\therefore \quad \frac{\lambda_1}{\lambda_4} = \frac{1}{4}$$

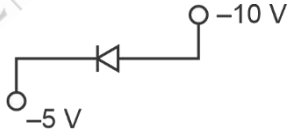
*** None of the option is matching.**

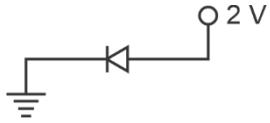
34. Which of the following circuits represents a forward biased diode?

(A) 

(B) 

(C) 

(D) 

(E) 

Choose the **correct** answer from the options given below:

(1) (B), (D) and (E) only (2) (A) and (D) only

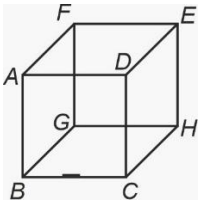
(3) (C) and (E) only (4) (B), (C) and (E) only

Answer (4)

Sol. For forward bias potential of p side should be higher than n side.



35. A line charge of length $\frac{'a'}{2}$ is kept at the center of an edge *BC* of a cube ABCDEFGH having edge length '*a*' as shown in the figure. If the density of line charge is λC per unit length, then the total electric flux through all the faces of the cube will be _____. (Take, ε_0 as the free space permittivity)



(1) $\frac{\lambda a}{16\varepsilon_0}$ (2) $\frac{\lambda a}{2\varepsilon_0}$

(3) $\frac{\lambda a}{4\varepsilon_0}$ (4) $\frac{\lambda a}{8\varepsilon_0}$

Answer (4)

Sol. Charge of the line charge = $\frac{a\lambda}{2}$

Portion of wire inside cube = $\frac{1}{4}$

$\therefore \quad q_{en} = \frac{1}{4}\left(\frac{a\lambda}{2}\right) = \frac{a\lambda}{8}$

$$\phi = \frac{q_{en}}{\varepsilon_0} = \frac{a\lambda}{8\varepsilon_0}$$

36. Given below are two statements :

Statement-I : The equivalent emf of two nonideal batteries connected in parallel is smaller than either of the two emfs.

Statement-II : The equivalent internal resistance of two nonideal batteries connected in parallel is smaller than the internal resistance of either of the two batteries.

In the light of the above statements, choose the correct answer from the options given below.

(1) Statement-I is false but Statement-II is true

(2) Statement-I is true but Statement-II is false

(3) Both Statement-I and Statement-II are true

(4) Both Statement-I and Statement-II are false

Answer (1)

Sol. In parallel connections

$$\frac{1}{r_{eq}} = \frac{1}{r_1} + \frac{1}{r_2}$$

$$\frac{E_{eq}}{r_{eq}} = \frac{E_1}{r_1} + \frac{E_2}{r_2}$$

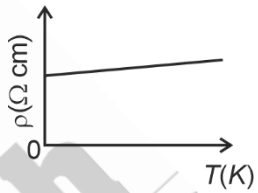
If $E_1 = E_2$ and $r_1 = r_2$, $E_{eq} = E_1 = E_2$

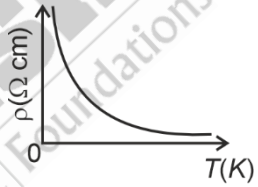
$\therefore$ Statement I is false.

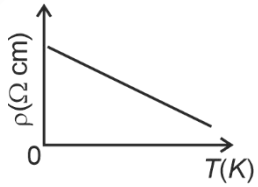
r_{eq} is less then both r_1 and r_2

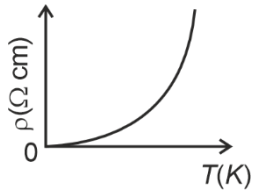
$\therefore$ Statement II is true

37. Which of the following resistivity (ρ) v/s temperature (T) curves is most suitable to be used in wire bound standard resistors?

(1)


(2)


(3)


(4)


Answer (1)

Sol. For bound standard resistors the resistivity will remain nearly constant.



38. A closed organ and an open organ tube are filled by two different gases having same bulk modulus but different densities ρ_1 and ρ_2, respectively. The frequency of 9th harmonic of closed tube is identical with 4th harmonic of open tube. If the length of the closed tube is 10 cm and the density ratio of the gases is $\rho_1 : \rho_2 = 1 : 16$, then the length of the open tube is :

(1) $\frac{15}{7}$ cm (2) $\frac{20}{9}$ cm

(3) $\frac{15}{9}$ cm (4) $\frac{20}{7}$ cm

Answer (2)

Sol. $9\left(\frac{v_c}{4l_c}\right) = 4\left(\frac{v_o}{2l_o}\right)$

$v = \sqrt{\frac{B}{\rho}}$

$\therefore \quad \frac{v_c}{v_o} = \sqrt{\frac{\rho_2}{\rho_1}} = 4$

$l_o = \frac{8}{9} l_c \left(\frac{v_o}{v_c}\right)$

$= \frac{8}{9} l_c \times \frac{1}{4} = \frac{2}{9} l_c$

$l_o = \frac{20}{9}$ cm

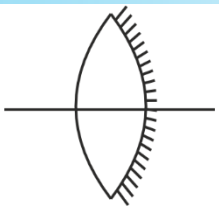
39. Given is a thin convex lens of glass (refractive index μ) and each side having radius of curvature R. One side is polished for complete reflection. At what distance from the lens, an object be placed on the optic axis so that the image gets formed on the object itself?

(1) R/μ (2) μR

(3) $R/(2\mu - 1)$ (4) $R/(2\mu - 3)$

Answer (3)

Sol.



$-\frac{1}{f_{eq}} = \frac{2}{f_l} - \frac{1}{f_m}$

$= 2(\mu - 1)\frac{2}{R} + \frac{2}{R}$

$-\frac{1}{f_{eq}} = \frac{2(2\mu - 1)}{R}$

$f_{eq} = -\frac{R}{2(2\mu - 1)}$

For concave mirror, object should be at $2f$ for the image to be at same point

Distance $= \frac{R}{(2\mu - 1)}$

40. Two spherical bodies of same materials having radii 0.2 m and 0.8 m are placed in same atmosphere. The temperature of the smaller body is 800 K and temperature of the bigger body is 400 K. If the energy radiated from the smaller body is E, the energy radiated from the bigger body is (assume, effect of the surrounding temperature to be negligible),

(1) E (2) 64 E

(3) 256 E (4) 16 E

Answer (1)

Sol. $P = \sigma\varepsilon A T^4$

$\frac{P_1}{P_2} = \frac{R_1^2 T_1^4}{R_2^2 T_2^4}$

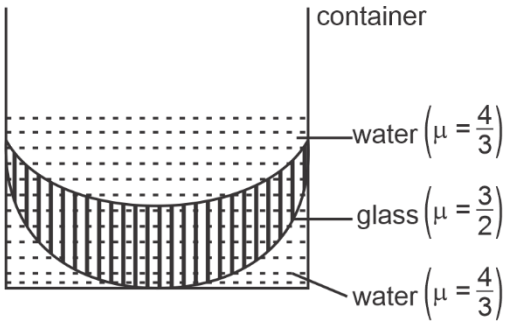
$\frac{P_1}{P_2} = \frac{1}{16} \times 2^4$

$P_1 = P_2$

$\therefore$ Energy will be same



41. In the diagram given below, there are three lenses formed. Considering negligible thickness of each of them as compared to $|R_1|$ and $|R_2|$, *i.e.*, the radii of curvature for upper and lower surface of the glass lens, the power of the combination is



(1) $\frac{1}{6}\left(\frac{1}{|R_1|}-\frac{1}{|R_2|}\right)$

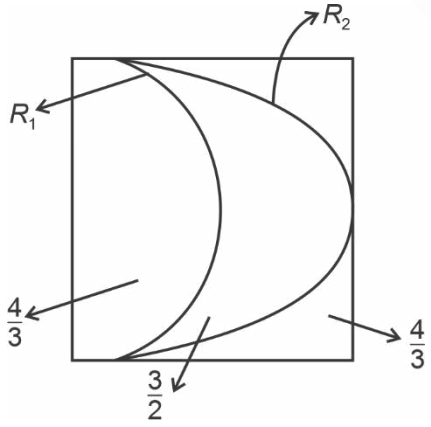
(2) $-\frac{1}{6}\left(\frac{1}{|R_1|}-\frac{1}{|R_2|}\right)$

(3) $-\frac{1}{6}\left(\frac{1}{|R_1|}+\frac{1}{|R_2|}\right)$

(4) $\frac{1}{6}\left(\frac{1}{|R_1|}+\frac{1}{|R_2|}\right)$

Answer (2)

Sol.



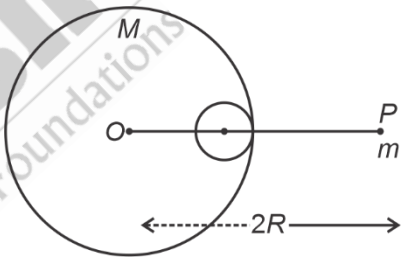
$$P_{eq}=P_1+P_2+P_3$$

$$=\left(\frac{4}{3}-1\right)\left(\frac{1}{R_1}\right)+\frac{1}{2}\left(-\frac{1}{R_1}+\frac{1}{R_2}\right)+\left(\frac{4}{3}-1\right)\left(-\frac{1}{R_2}\right)$$

$$=\frac{1}{3}\left(\frac{1}{R_1}-\frac{1}{R_2}\right)-\frac{1}{2}\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

$$P_{eq}=-\frac{1}{6}\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

42. A small point of mass m is placed at a distance $2R$ from the centre 'O' of a big uniform solid sphere of mass M and radius R. The gravitational force on 'm' due to M is F_1. A spherical part of radius $\frac{R}{3}$ is removed from the big sphere as shown in the figure and the gravitational force on m due to remaining part of M is found to be F_2. The value of ratio $F_1 : F_2$ is



(1) 12 : 9

(2) 11 : 10

(3) 12 : 11

(4) 16 : 9

Answer (3)

Sol. $F_1=\frac{GMm}{4R^2}$

$$F_2=\frac{GMm}{4R^2}-G\left(\frac{M}{27}\right)\frac{m}{\left(\frac{4R}{3}\right)^2}$$



$$= \frac{GMm}{R^2}\left(\frac{1}{4} - \frac{1}{48}\right)$$

$$= \frac{11}{48}\ \frac{GMm}{R^2}$$

$$\therefore \quad \frac{F_1}{F_2} = \frac{12}{11}$$

43. Given below are two statements :

Statement-I: In a vernier callipers, one vernier scale division is always smaller than one main scale division.

Statement-II: The vernier constant is given by one main scale division multiplied by the number of vernier scale divisions.

In the light of the above statements, choose the correct answer from the options given below.

(1) Both Statement-I and Statement-II are true

(2) Both Statement-I and Statement-II are false

(3) Statement-I is false but Statement-II is true

(4) Statement-I is true but Statement-II is false

Answer (2)

Sol. 1 vernier scale division is not always smaller than 1 main scale divisions statement-I is not correct.

Vernier constant = 1 main scale division – 1 vernier scale division

Statement-II is not true

44. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion-(A): If Young's double slit experiment is performed in an optically denser medium than air, then the consecutive fringes come closer.

Reason-(R): The speed of light reduces in an optically denser medium than air while its frequency does not change.

In the light of the above statements, choose the **most appropriate answer** from the options given below :

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

(2) **(A)** is true but **(R)** is false

(3) Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**

(4) **(A)** is false but **(R)** is true

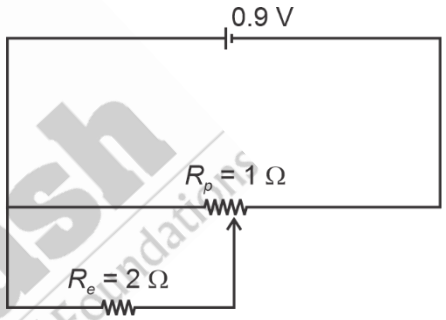
Answer (1)

Sol. $\beta = \frac{\lambda D}{d}$

In denser medium $\lambda' = \frac{\lambda}{\mu}$

$\therefore$ β decreases.

45.



Sliding contact of a potentiometer is in the middle of the potentiometer wire having resistance R_p = 1 Ω as shown in the figure. An external resistance of R_e = 2 Ω as connected via the sliding contact. The electric current in the circuit is :

(1) 1.35 A

(2) 1.0 A

(3) 0.3 A

(4) 0.9 A

Answer (2)



Sol. $R_{eq} = (0.5\ \Omega \,||\, 2\ \Omega) + 0.5\ \Omega$

$$= \frac{2}{5} + \frac{1}{2} = \frac{9}{10}\ \Omega$$

$$I = \frac{V}{R_{eq}} = \frac{0.9}{0.9} = 1\ \text{A}$$

SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

46. The position vectors of two 1 kg particles, (A) and (B), are given by $\vec{r}_A = \left(\alpha_1 t^2\,\hat{i} + \alpha_2 t\,\hat{j} + \alpha_3 t\,\hat{k}\right)$ m and $\vec{r}_B = \left(\beta_1 t\,\hat{i} + \beta_2 t^2\,\hat{j} + \beta_3 t\,\hat{k}\right)$ m, respectively; (α_1 = 1 m/s^2, $\alpha_2 = 3n$ m/s^2, α_3 = 2 m/s, β_1 = 2 m/s, β_2 = –1 m/s^2, $\beta_3 = 4p$ m/s), where t is time, n and p are constants. At t = 1 s, $|\vec{V}_A| = |\vec{V}_B|$ and velocities $\vec{V}_A$ and $\vec{V}_B$ of the particles are orthogonal to each other. At t = 1 s, the magnitude of angular momentum of particle (A) with respect to the position of particle (B) is $\sqrt{L}$ kgm^{2}s^{-1}. The value of L is _____.

Answer (90)

Sol. At $t = 1$

$$r_{AB} = -1\hat{i} + (3n+1)\hat{j} + (2-4p)\hat{k}$$

At $t = 1$

$$v_A = 2\hat{i} + 3n\,\hat{j} + 2\hat{k}$$

$$v_B = 2\hat{i} - 2\,\hat{j} + 4p\hat{k}$$

$$\vec{v}_A \cdot \vec{v}_B = 0, \quad 4 - 6n + 8p = 0$$

$$|v_A| = |v_B| \qquad (3n)^2 + 4 = 4 + 16p^2$$

$$3n = -4p$$

$$4 + 16p = 0$$

$$p = -\frac{1}{4},\ n = \frac{1}{3}$$

$$r_{AB} = -\hat{i} + 2\hat{j} + 3\hat{k}$$

$$v_A = 2\hat{i} + \hat{j} + 2\hat{k}$$

$$\therefore\ L = m\left|\vec{r}_{AB} \times \vec{v}_A\right| = 90$$

47. The driver sitting inside a parked car is watching vehicles approaching from behind with the help of his side view mirror, which is a convex mirror with radius of curvature R = 2 m. Another car approaches him from behind with a uniform speed of 90 km/hr. When the car is at a distance of 24 m from him, the magnitude of the acceleration of the image of the car in the side view mirror is 'a'. The value of 100a is _____ m/s^2.

Answer (8)

Sol. $\frac{1}{V} + \frac{1}{U} = \frac{1}{f}$

$$\frac{1}{V} - \frac{1}{24} = 1$$

$$V = \frac{24}{25}$$

$$m = -\frac{V}{U} = \frac{1}{25}$$

$$|V_{\text{image}}| = m^2\, V_{\text{object}} = \frac{1}{25}\ \text{m/s}$$



$$\frac{1}{V}+\frac{1}{U}=\frac{1}{f}$$

$$\frac{d^2V}{dt^2}=V^2\left[\frac{2}{V^3}\left(\frac{dV}{dt}\right)^2+\frac{2}{U^3}\left(\frac{dU}{dt}\right)^2\right]$$

$$a=-\frac{2}{25}$$

$$|100a|=8$$

48. A particle is projected at an angle of 30° from horizontal at a speed of 60 m/s. The height traversed by the particle in the first second is h_0 and height traversed in the last second, before it reaches the maximum height, is h_1. The ratio $h_0 : h_1$ is _____.

[Take, g = 10 m/s^2]

Answer (5)

Sol. U = 60 m/s

θ = 30°

$$t_{H_{max}}=\frac{U\sin\theta}{g}=\frac{60}{10}\times\frac{1}{2}=3\text{ s}$$

Height in 1st second $= 30\times1-\frac{1}{2}\times10\times1$

$= 25$ m

Height in last second = 5 m

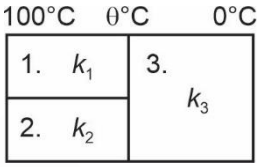
$$\frac{25}{5}=5$$

49. Two soap bubbles of radius 2 cm and 4 cm, respectively, are in contact with each other. The radius of curvature of the common surface, in cm, is _____.

Answer (4)

Sol. $R=\frac{R_2R_1}{R_2-R_1}=\frac{4\times2}{2}=4$ cm

50. Three conductors of same length having thermal conductivity k_1, k_2 and k_3 are connected as shown in figure.



Area of cross sections of 1st and 2nd conductor are same and for 3rd conductor it is double of the 1st conductor. The temperature are given in the figure. In steady state condition, the value of θ is _____ °C.

(Given : k_1 = 60 Js^{-1}m^{-1}K^{-1}, k_2 = 120 Js^{-1}m^{-1}K^{-1}, k_3 = 135 Js^{-1}m^{-1}K^{-1})

Answer (40)

Sol. $R_1=\frac{2\ell}{k_1A}$, $R_2=\frac{2\ell}{k_2A}$, $R_3=\frac{\ell}{k_3A}$

$$(100-\theta)\left(\frac{1}{R_1}+\frac{1}{R_2}\right)=\frac{(\theta-0)}{R_2}$$

$$(100-\theta)\frac{(k_1+k_2)A}{2\ell}=\frac{\theta k_3A}{\ell}$$

$$(100-\theta)(k_1+k_2)=2\theta k_3$$

$$\theta=\frac{100(k_1+k_2)}{k_1+k_2+2k_3}$$

$$=\frac{100\times180}{450}=40$$



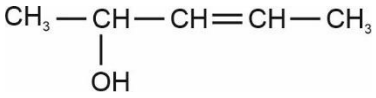
CHEMISTRY

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

51. How many different stereoisomers are possible for the given molecule?



(1) 2 (2) 4

(3) 3 (4) 1

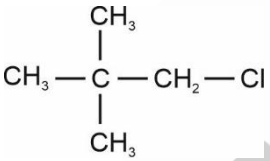
Answer (2)

Sol. n = 2

Number of stereoisomers = $(2)^2 = 4$

52. Given below are two statements:

Statement I : $CH_3 - O - CH_2 - Cl$ will undergo S_N1 reaction though it is a primary halide.

Statement II :  will not undergo S_N2 reaction very easily though it is a primary halide.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **Statement I** and **Statement II** are incorrect

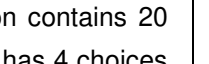
(2) Both **Statement I** and **Statement II** are correct

(3) **Statement I** is correct but **Statement II** is incorrect

(4) **Statement I** is incorrect but **Statement II** is correct

Answer (2)

Sol. Both statement-I and statement-II are correct

Statement – I :  is highly stable carbocation.

Statement – II : Due to hindrance, S_N2 will not take place easily.

53. A liquid when kept inside a thermally insulated closed vessel at 25ºC was mechanically stirred from outside. What will be the correct option for the following thermodynamic parameters?

(1) $\Delta U > 0, q = 0, w > 0$ (2) $\Delta U < 0, q = 0, w > 0$

(3) $\Delta U = 0, q < 0, w > 0$ (4) $\Delta U = 0, q = 0, w = 0$

Answer (1)

Sol. $q = 0$

$w > 0$

$\Delta U > 0$

54. In which of the following complexes the CFSE, Δ_0 will be equal to zero?

(1) $K_3[Fe(SCN)_6]$ (2) $[Fe(en)_3]Cl_3$

(3) $[Fe(NH_3)_6]Br_2$ (4) $K_4[Fe(CN)_6]$

Answer (1)

Sol. $K_3[Fe(SCN)_6]$ will have CFSE equal to zero (Fe^{3+} : $4s^0 3d^5 \left(t_{2g}^3 e_g^2\right)$

55. Which of the following electronegativity order is **incorrect**?

(1) Mg < Be < B < N (2) Al < Si < C < N

(3) Al < Mg < B < N (4) S < Cl < O < F

Answer (3)

Sol. Electronegativity of Al is higher than that of Mg

$\Rightarrow$ option (3) is correct

